

Ketan Mhaske

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Exam	University	Institute	Year	CPI/%
Graduation	D.batu University	Deogiri Institute of Enginnering And Management Studies,chh. SambhajiNagar (Aurangabad)	2019-23	8.0
Intermediate/+2	HSC	Samarth Junior Collage of Science ,Arts & Commerce ,chh.SambhajiNagar	2017-19	69.7
Matriculation	SSC	Swami Vivekanand Academy, chh.SambhajiNagar (Aurangabad)	2017	78.60

EXPERIENCE

Software Developer

Nov 2023 – Present

MicroWorld Software Services Ltd Pvt

Andheri, Mumbai

- Led the development of performance-critical software applications in C and C++, with a focus on algorithm optimization and system efficiency.
- Designed and implemented cross-platform C++ solutions using the Qt framework, enhancing application usability on Linux and macOS.
- Developed a Linux kernel module to intercept system calls, enabling custom monitoring and control of file operations based on ransomware file extensions.
- Contributed to a vulnerability detection tool integrating C/C++, SQLite3, and Bash, improving security tracking and system resilience.
- Improved system performance by optimizing low-level code and integrating efficient file handling routines in C++.
- Gained exposure to Linux kernel development and basic security mechanisms through the creation of system monitoring tools.
- Implemented systems for retrieving and deploying updated policy and virus definitions to client systems, improving security efficiency.
- Applied fundamental knowledge of system-level performance optimization, basic networking, and security protocols.

PROJECTS

Cross-Platform USB Storage Encryption Module | C, Linux Kernel, RC4, Kprobes, Block I/O

- Developed The fully dynamic auto usb encryption/decryption kernel module at full system level coding .
- Developed production-grade kernel module providing transparent USB MBR/VBR encryption using RC4 stream cipher with Windows driver compatibility.
- Implemented cross-version compatibility layer supporting 15+ kernel versions (5.4-6.12+) across Ubuntu 20.04-24.04, RHEL/CentOS 8-9.
- Utilized kprobes for I/O interception at bio layer, intelligent operation detection differentiating format operations from user data writes.
- Working/Usage - encrypt the all usb deviceess which trying to copy data from system and then that device unsable without our module user cant use that util format.

Device Control Daemon for Linux | C++, Linux Kernel, Udev, libudev, Qt

- Developed a device control daemon service in C++ for Linux, managing various device connections such as USB, CD-ROM, SD cards, Bluetooth, and Wi-Fi.
- Enabled user-defined control over blocking or allowing specific devices through a user-friendly interface using Qt.
- Utilized udev and libudev libraries alongside custom kernel modules to intercept and manage device connections across Linux distributions.
- Implemented robust security measures to protect the system from unauthorized device access while allowing flexible user configurations.

Two-Factor Authentication for Linux (2FA) | C++, PAM, Google Authenticator, Qt

- Developed a 2FA system in C++ for Linux user login (GDM), SSH, and other authentication mechanisms, providing OTP-based security.

- Implemented flexibility for applying 2FA to specific users or all users, with the OTP generated every 30 seconds on the user's mobile device.
- Integrated Google Authenticator library and PAM modules to provide a secure, customizable authentication system.
- Created an intuitive UI using Qt to allow easy configuration and management of 2FA settings by system administrators.

Website Blocking Daemon Service | *C++, Linux Kernel, nfeque, Packet Capture*

- Contributed to a website blocking daemon service using C++ and the nfeque library to capture and inspect network packets.
- Developed a kernel module for capturing and extracting packet data, enabling the blocking of websites based on user-defined criteria.
- Implemented efficient packet inspection and filtering mechanisms to block unwanted websites and enhance user browsing security.

libsinc – Cross-Platform Multi-Cloud Directory Sync Library | *C/C++, Linux, Threads, libcurl, jsoncpp*

- Designed and implemented a modular C/C++ dynamic library for Linux to sync a target directory with OneDrive, Google Drive, and Dropbox.
- Launched separate threads for each provider to enable parallel and policy-driven syncing every 10 minutes.
- Utilized access and refresh tokens, along with client credentials, to securely obtain and renew cloud-specific tokens via REST APIs.
- Integrated libcurl for HTTP interactions and jsoncpp for efficient parsing of API responses.
- Automatically creates the desired folder on the cloud (if missing) and avoids duplicate folder or file uploads.
- Implements bi-directional sync logic: deleted files from the cloud are re-uploaded locally, and deleted local files are removed from the cloud.
- Engineered for reusability and integration—exposes clean API endpoints via a shared library interface.

Attachment Control System for Linux | *C/C++, inotify, fanotify, libpcap, TLS, SNI, Socket Inspection*

- Built a file access control system using inotify and fanotify to monitor file events like open, access, and read in real-time.
- Captured process information (PID, binary name) for actions by browsers or tools like Firefox, Chrome, curl, etc.
- Implemented policy-based logic to allow or deny file attachments based on process behavior and configuration.
- Extended functionality to inspect TLS handshakes using libpcap, retrieving SNI and IP details of sites accessing local files.
- Logged SNI, IP, socket, and PID information to correlate file access with browser sessions.
- Built a socket inspection module to analyze active IPs and sockets, identifying if a browser is uploading the file.
- Blocked or allowed file transfers to specific websites based on domain, socket activity, and configured policy.

TECHNICAL SKILLS

Languages: C/C++, Bash, Basic Java, HTML, CSS, JS

Frameworks & APIs: Qt, POSIX, PAM, REST APIs, OAuth 2.0

Operating Systems: Linux, macOS

Developer Tools: Git, Vim, GDB, Valgrind, Make, CMake

Libraries: boost, cJSON, curl, spdlog, jsoncpp

Systems Programming: Multithreading, Socket Programming, Linux Kernel, Netfilter, Device Management

Networking/Security: Packet Capture, Regex, 2FA Integration, File Monitoring

Soft Skills: DSA, OOPs, Design Patterns, Problem Solving